

Wenzhi Lyu

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EDUCATION

Northwestern University (Dual Master's Degree)

Jan 2025 – Dec 2025

M.S. in Social and Economic Policy (GPA: 3.85/4.0)

- **Selected Coursework:** Graduate-Level Macroeconomic Modeling (A), NLP and LLM (A), Public Finance, Causal Methods, Economics of Public Policy, Economics of Crime

The Chinese University of Hong Kong (Dual Master's Degree)

Aug 2024 – Jul 2026

M.S. in Applied Economics (GPA: 3.77/4.0)

- **Selected Coursework:** Graduate-Level Macroeconomics (A), Microeconomics, Econometrics, Mathematical Economics, Machine Learning (A), Data Science (A)

Xi'an Jiaotong University

Aug 2019 – Jul 2023

Bachelors of Economics (GPA: 3.39/4.3); Bachelors of Management (GPA: 3.73/4.3)

- **Selected Coursework:** Probability Theory (A), Mathematical Modeling (A), Statistics (A), Python (A), Intermediate Macroeconomics, Microeconomics, Econometrics, Calculus, Linear Algebra

RESEARCH INTERESTS

Urban Economics; Macroeconomics; Spatial Economics; Technological Change and AI

WORKS IN PROGRESS

The Political-Economy of China's Industrial and TFP Spatial Variance: A Dynamic Spatial General Equilibrium Approach

Aug 2024 – Present

Co-author: Zhongji Wei

- Conceiving and deriving a dynamic spatial general equilibrium model of heterogeneous firms and local governments with the coauthor.
- Conducting large-scale firm-level data collection and cleaning, and using Matlab and Stata for model calibration, parameter estimation, and counterfactual policy analysis.
- Using the model to quantify how local policy instruments affect firm production and to run welfare, efficiency, and counterfactual policy experiments that identify key drivers of cross-regional TFP differences.

The Robots are Coming for Our Safety: Automation's Impact on U.S. Crime Rates

Solo Author

Sept 2025 – Present

- Built a 2001–2014 Commuting Zone-level panel (FBI crime, Census, IFR robots) and used Stata to clean microdata, construct automation-exposure measures, and merge them to crime outcomes.
- Estimated 2SLS shift-share regressions and counterfactual scenarios showing robot exposure raises property crime but not violent crime, and ran mechanism “horse-race” and pre-period/spatial placebo tests.

The Robots are Coming for Our Homes: Automation's Impact on the U.S. Housing Market

Solo Author

Sept 2025 – Present

- Built a 2001–2019 county-level panel of U.S. housing markets and automation exposure (FHFA, Census, IFR), used Stata to clean microdata and construct housing and robot variables, and aggregated them to Commuting Zones via population-weighted averages.
- Estimated 2SLS shift-share regressions and counterfactuals to quantify robot exposure's impact on rents, prices, and homeownership, ran mechanism and robustness tests, and found automation lowers rents, reduces homeownership, and widens regional inequality.

RESEARCH EXPERIENCE

University of North Carolina, Kenan-Flagler Business School

Evanston, USA (Remote)

Research Assistant, Supervised by Prof. Franklin Qian & Caitlin Gorbach

Mar 2025 – Present

- Assisted with the empirical workflow for the paper “The Impact of Institutional Owners on Housing Markets” under the supervision of Professors, processing large-scale deed recordings and administrative datasets.
- Helped maintain and refine the existing Stata & Python codebase for data construction and empirical analysis by checking outputs, debugging code, and implementing improvements with another RA.

- Assisted in implementing the project's 2SLS identification strategy, including improving the code of the shift-share instrument that combines the "Suitability Index" with the "Declining Management Costs".
- Ran assigned econometric analyses—such as temporal heterogeneity, pre-period placebo, and mechanism exercises—helped validate results and organize the replication package under faculty supervision.

Harvard University, Harvard Business School

Hong Kong SAR (Remote)

Research Assistant, Supervised by Prof. Letian Zhang

Oct 2024 – Apr 2025

- Collected, cleaned, and organized the global leadership database in Stata and Python.
- Analyzed the data and summarized empirical findings for research reports, papers, and presentations.

The Chinese University of Hong Kong

Hong Kong SAR, China

Research Assistant, Supervised by Prof. Ce Shi

Feb 2025 – Jun 2025

- Collected, cleaned, and classified electric vehicle industry data from multiple sources, and conducted quantitative analysis and visualization in Stata/Python.
- Compiled literature reviews through systematic searches and contextual mapping, and assisted in drafting research reports/publications.

Camphor Economics Young Scholar Summer School

Beijing, China

Organizer: University of Chinese Academy of Social Sciences

Jul 2024 – Aug 2024

- Selected from 357 applicants (10% acceptance) for intensive training in economic research, econometrics, and academic writing with leading scholars and peers, strengthening my research design skills.

The Impact of Third-Party Payment on The Profitability of Commercial Banks in China (Graduation Thesis)

Xi'an, China

Sept 2022 – Jul 2023

Advisor: Prof. Shen Gu

- Reviewed literature on corporate finance and commercial banking and used fixed-effects and mediation models to show that the rapid expansion of third-party payment institutions reduces Chinese banks' profitability and cash flow, with liquidity and net interest margin as key mediating channels.

The Impact of Executive Monetary Compensation on Corporate Performance: Evidence From The A-Share Market (Graduation Thesis)

Xi'an, China

Sept 2022 – Jul 2023

Advisor: Prof. Xin Wu

- Analyzed A-share listed firms using fixed-effects and mediation models, finding that higher executive cash compensation reduces profitability and cash flow and crowds out innovation by tightening financial constraints.

INTERNSHIP EXPERIENCE

Guotai Junan Securities

Industrial Research

Shanghai, China

Mar 2023 – Jul 2023

- Assisted analysts in writing an in-depth report on six A-share white-feather broiler firms, built net asset value models for five target companies, and prepared data/materials and summary reports for investor roadshows.

China Golden Travel Investment Holding Co., Ltd.

Fund Intern

Xi'an, China

Jul 2022 – Aug 2022

- Collected and analyzed data for private equity investment projects, drafted investment reports, tracked industry policies, assisted across the investment process, and developed a simulated investment strategy report and code.

HONORS AND AWARDS

- Performance-Based Scholarships (\$15,000 HKD) in Course: Machine Learning Methods for Economists, CUHK
- Performance-Based Scholarships (\$15,000 HKD) in Course: Data Science for Economists, CUHK
- Excellent Award for Entrepreneurship and Innovation Track in the 33rd Xi'an Jiaotong University Tengfei Cup Innovation Competition
- Summer Social Practice Excellence Award of Qian Xuesen College, XJTU
- Outstanding Officer of the Student Union of Qide College, XJTU

TECHNOLOGIES AND LANGUAGES

Languages Skills: English (IELTS 7), Mandarin (native)

Technologies: Python, R, Stata, L^AT_EX, Matlab